

TABLE II
A Comparison: Select General-Purpose, Mid-Range Multimeters
in the Market Today (Indian Brands)

Brand	Model	Specifications							True RMS	Safety	Display
		Range									
		Volt (AC) *	Volt (DC)	Resistance	Capacitance	Diode	Current (AC)	Current (DC)			
HTC Instruments	DM-87	0.1mV - 750V	0.1mV - 1000V	0.1Ω - 30MΩ	10pF - 2000μF	N/A	0.1μA - 20A	0.1μA - 20A	-	-	4½ digits, LCD display
Kusam-MECO	KM 629	999.9mV/ 9.999V/ 99.99V/ 750V	999.9mV/ 9.999V/ 99.99V/ 999.9V	0.1Ω - 40MΩ	1μF/10μF/ 100μF/ 1mF/10mF	2V	40μA/40mA/ 10A	40μA/400μA/ 4mA/40mA/ 4A/10A	Yes	EC1010-1 (1995), EN61010-1 (1995), UL311 1-1 (6.1994), CSAC22.2 No. 1010-1-92 to terminals	Four digits, 9999 counts, LCD display
MetroQ	MTQ 70A	400mV/4V/ 40V/400V/ 750V	400mV/ 4V/40V/ 400V/ 1000V	400Ω/4KΩ/ 40KΩ/ 400KΩ/ 4MΩ/40MΩ	4nF/40nF/ 400nF/ 4μF/40μF/ 200μF	Available	30mA/300mA/ 10A	30mA/ 300mA/10A	Yes	CAT-IV	3½, LCD display
Mextech Technologies	DT-115	0.1mV - 750V	0.1mV - 1000V	0.1Ω - 40MΩ	40nF/400nF/ 4mF/40mF/ 100mF	Available	0.1mA - 10A	0.1mA - 10A	Yes	-	LCD display
Motwane	M 63	60mV - 600mV/ 6V - 750V	60mV - 600mV/ 600mV - 1000V	600Ω - 60MΩ	40nF - 4000μF	Available	60mA - 20A	60mA - 20A	Yes	CAT IV/600V, IP 54, CE	LCD display
Salicon	ST82	200mV/2V/ 20V/200V/ 750V	200mV/2V/ 20V/200V/ 1000V	200/2k/ 2k/200k/ 2M/20MΩ	-	Available	200μA/2000μA/ 20mA/200mA/ 20A	200μA/ 2000μA/ 20mA/200mA/ 20A	-	-	LCD display
Scientific	SM 7022	1mV - 750V	0.1mV - 1000V	0.1Ω - 40MΩ	1pF - 200μF	Available	1μA - 10A	1μA - 10A	-	-	3½-digit LCD display, 4000 counts

applications in universities and research and development labs.

Detachable display. This feature enables an engineer to take a clamp-on reading, be it voltage, current, resistance or capacitance, by making use of wireless technology and display of the multimeter, which is detachable. The user can record data from up to 9.1m (30-feet) away from the loca-



Fig. 3: Fluke's 233 in action

tion of the test probes and meter body. This feature can be seen in Fluke's 233 wireless digital multimeter.

Make the right choice

Following parameters could help you in locking

down the multimeter perfect for your usage.

Safety comes first. A digital multimeter has the capability to identify a problem in a circuit, but it is important to know its limitations else it could result in equipment damage and possibly cause personal injury.

Before buying one, you must ensure that it is designed to pass international standards of electrical safety, such as CAT (a safety standard for electrical installations and equipment). For single-phase industrial use and three-phase distribution,



Fig. 4: Different standards for digital multimeters (Image courtesy: Fluke)